Intelligent GIS-Based Urban Services Platform in Astana

An AI-enhanced, GIS-based platform that integrates **parking availability and reservation, hazard reporting, lost-and-found search and a decision-support map for city staff** on one GIS data model for Astana, with three AI modules: hazard-report classification, lost-item matching and 30-minute parking prediction.

> **Status:** analysis, design and planning stage (milestone 24 October 2026). The repository documents analysis, design, planning and supervision and contains wireframe prototype materials. The full application is not implemented yet.

Team

Name	Role	Group
Arlan Aidarov (Айдаров Арлан Адилханулы)	Developer	SE-2412
Ramazan Ashirbek (Аширбек Рамазан Толегенулы)	Developer	SE-2412
Kairat Bulatov (Булатов Кайрат Булатович)	Developer	SE-2412
Supervisor	Algys Saltanat	

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Problem and aim

City data stay in separate systems, so urban services work as isolated tools. No study found builds and evaluates one GIS data model that serves parking and reservation, hazard reporting and lost-and-found with AI support for a specific city such as Astana. Aim: design, implement and evaluate a prototype and test whether it beats baselines (keyword rules, text filter, historical average, baseline workflow) on technical and user measures. See `pdf/Midterm_Research_Proposal.pdf` and the documents below.

Functions and AI modules

Function	AI module
Parking availability and reservation	30-minute occupancy prediction (time-series regression)
Hazard reporting (KZ/RU/EN, photo, GPS)	Report classification (multilingual text encoder + classifier)
Lost-and-found search	Matching (image embeddings + text + location), top-5
Decision-support map for staff	– (aggregates all layers)

Out of scope: live deployment with the city, other domains (energy, utilities), full digital twin, payments, navigation.

Planned technology

PostgreSQL + PostGIS · Python REST API · web map interface on OpenStreetMap (Leaflet) · Python ML libraries · synthetic data generators · Docker Compose. Justification: `docs/05-technology-comparison.md`.

Documents

All documents exist as Markdown in `docs/` and as PDF in **`pdf/`**. | # | Document | |---|---| | – | Midterm Research Proposal (`pdf/Midterm_Research_Proposal.pdf`) | | 1 | Project proposal and scope (`docs/01-project-proposal.md`) | | 2 | Evidence of relevance (`docs/02-evidence-of-relevance.md`) | | 3 | Literature review (`docs/03-literature-review.md`) | | 4 | Existing system comparison (`docs/04-existing-system-comparison.md`) | | 5 | Technology comparison (`docs/05-technology-comparison.md`) | | 6 | Requirements specification (`docs/06-requirements-specification.md`) | | 7 | Architecture and diagrams (`docs/07-architecture.md`) | | 9 | Verification and evaluation plan (`docs/09-verification-and-evaluation-plan.md`) | | 11 | Supervisor feedback (`docs/11-supervisor-feedback.md`) | | – | YouTrack backlog, contributions (`docs/project-management/`) | | – | Checklist for 24 October (`CHECKLIST_24_OCT.md`) |

Diagrams

Editable sources: `diagrams/src` (Mermaid `.mmd`, Graphviz `.dot`); images: `diagrams/png` — system context, use cases, component diagram (draft), data model, hazard-report flow, reservation flow.

Prototype

Wireframes of five screens: `prototype/` (SVG sources + PNG + `pdf/Prototype-wireframes.pdf`).

Project management

YouTrack: **`<add link>`** — 38 prioritised tasks with assignees (CSV import in `docs/project-management/youtrack-import.csv`).

Repository structure

```
├── README.md
├── CHECKLIST_24_OCT.md
├── docs/        documents (Markdown), evidence/, project-management/
├── diagrams/    src/ (editable) and png/
├── prototype/   wireframes (src/ SVG, png/)
├── pdf/         all documents as PDF + Midterm Research Proposal
└── (src/, tests/, data/, models/ — added during implementation)
```

Work plan

Weeks 1–3 requirements · 4–8 database, API, map interface · 5–9 AI modules · 10–12 user study · 13–14 analysis and article.

Licence

(choose after checking the licences of models and datasets used)